

CS 2160: Computer Organization & Assembly Language Programming

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Assignment #01

- This assignment is worth **100 points**.
- Please type your submissions and format properly (the TA will thank you)
- **10% bonus points** will be available for anyone submitting LaTeX-formatted homework
- Please use the internet and all other resources to help you with the assignment, not write it for you

Problem 01 ($3 \times 5 = 15$ pts)

Take a look at your (or someone's) *personal computer* (laptop, desktop, etc.) and find out the following information:

1. What *processor* is being used there?
2. What is the *clock speed* of the processor?
3. How many *cores* does the processor have?
4. What is the *capacity* of hard drive?
5. What *operating system* does it have?

Note: can you find out the answer(s) without looking at the actual hardware or looking up the answers online?

Problem 02 ($5 \times 2 = 10$ pts)

Search up the following questions and then answer in your own words:

1. What is **Moore's law**? Does it still hold?
2. What are **memory walls** and **power walls** in computer architecture? How did they affect computer design?

Problem 03 (25 pts)

Translate the following **C** code into **RISC-V Assembly**. Assume registers **t3** and **t4** are associated with integer variables **i** and **j** respectively. Assume the base address of arrays **A** and **B** are in registers **a0** and **a1** respectively.

```
#include <stdio.h>
int main() {
    // arbitrary C code

    B[5] = A[i] + A[j];

    return 0;
}
```

Problem 04 (25 pts)

For the **RISC-V Assembly** instruction below, what is the corresponding **C** code? Assume registers **t0**, **t1** are associated with integer **f**, **g** respectively. Assume the base address of integer arrays **A** and **B** are in registers **a0** and **a1** respectively. Also assume that registers **t2-t6** are available to use.

```
slli t2, t0, 2
add t2, a0, t2
lw t0, 0(t2)
lw t2, 4(t2)
add t2, t2, t0

slli t3, t1, 2
add t3, a1, t3
sw t2, 0(t3)
```

Problem 05 (25 pts)

Suppose register **t0** holds the integer value 85,001 and the register **a0** holds the value 0x0100.

```
sw t0, 0a(0)
```

1. Assume the computer uses **little-endian** format. What is the data that is held in memory addresses 0x0100, 0x0101, 0x0102, 0x0103? Write your answer in *hexadecimal*.
2. Same as the previous problem, but this time assume the computer uses **big-endian** format.
3. Let's again assume **little-endian** format. Consider the **Assembly** code:

```
sw t0, 0(a0)
srli t0, t0, 12
sb t0, 2(a0)
```

What is the data that is held in memory addresses 0x0100, 0x0101, 0x0103 after this code is run? Write your answer in *hexadecimal*.